CS24307 — THEORY (3.0 CR) | MODULE V NOTES | BIT MESRA | NEP SCHEME

Module 5: Machine Learning Foundations

Inductive Learning, ID3 Decision Trees (Entropy/Gain), Perceptrons & Backpropagation

Module V: Machine Learning & Neural Networks — Complete 9-Topic Syllabus Tracker

Topic 31: What is Learning? (Experience to Performance Improvement)

Topic 32: Rote Learning (Direct Memorization & Caching)

Topic 33: Learning by Taking Advice (Expert Instruction Parsing)

Topic 34: Learning from Examples (Inductive Supervised Classification)

Topic 35: Induction vs. Deduction in Machine Learning

Topic 36: Formal Learning Theory (Hypothesis Space, Bias & Variance)

Topic 37: Neural Net Learning (Perceptrons, Activations & Backpropagation)

Topic 38: Underfitting (High Bias, Low Capacity Models)

Topic 39: Overfitting (High Variance, Noise Memorization & Regularization)

Topic 31 – 35: Forms of Machine Learning & Inductive Inference

Learning Paradigm	Operational Mechanism	Key Advantages & Tradeoffs
1. Rote Learning	Stores computed results and solutions directly in a lookup table / cache. Re-executing retrieves pre-computed solutions.	Instant $O(1)$ retrieval; zero generalization capability to unseen problem instances.
2. Learning by Advice	An external teacher/expert provides high-level instructions, which the system translates into internal operational rules.	Rapid knowledge bootstrapping; requires complex natural language understanding.
3. Learning from Examples	Induces general classification/prediction functions $y = f(x)$ from a set of labeled input-output pairs $\langle x_i, y_i \rangle$.	The foundation of modern machine learning; requires large representative datasets.
4. Induction vs. Deduction	• **Induction:** Derives general hypotheses from specific observations (probabilistic). • **Deduction:** Derives specific consequences from general axioms (guaranteed true).	Induction discovers new knowledge from data; Deduction proves theorems from axioms.

Topic 36: Formal Learning Theory (Hypothesis Space & Bias-Variance)

A learning algorithm searches through a **Hypothesis Space** $\mathcal{H}$ to select a candidate hypothesis $h \in \mathcal{H}$ that approximates the true underlying target function $f(x)$ with minimal generalization error on unseen data.

Concept	Mathematical / Theoretical Definition
Bias	Error from erroneous assumptions in the learning algorithm (e.g., assuming linear boundary for nonlinear data).
Variance	Error from sensitivity to small fluctuations in the training set (model memorizes random training noise).
Generalization	The ability of a trained model to accurately predict labels on novel, previously unseen test samples.

Topic 37: Neural Network Learning & Gradient Descent

1. Artificial Neuron (Perceptron) Formulation:

$$z = \sum_{i=1}^{n} w_i x_i + b = \mathbf{w}^T \mathbf{x} + b$$

$$y = f(z) \quad \text{(Activation Function)}$$

- **Sigmoid:** $\sigma(z) = \frac{1}{1+e^{-z}} \in (0, 1)$ - **ReLU (Rectified Linear Unit):** $\text{ReLU}(z) = \max(0, z)$ - **Tanh:** $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \in (-1, 1)$

2. Gradient Descent & Backpropagation Weight Update:

$$w_{ij} \leftarrow w_{ij} - \eta \frac{\partial \mathcal{L}}{\partial w_{ij}}$$

Where $\eta > 0$ is the learning rate and $\mathcal{L}$ is the Mean Squared Error loss function $\mathcal{L} = \frac{1}{2} \sum (t_k - y_k)^2$.

Topic 38 & 39: Underfitting vs. Overfitting

Characteristic	Underfitting (High Bias)	Overfitting (High Variance)
Root Cause	Model is too simplistic with insufficient expressive capacity.	Model is excessively complex with too many parameters.
Training Error	**High** (cannot fit training data).	**Ultra-Low / Zero** (memorizes training noise).
Testing / Validation Error	**High** (poor performance everywhere).	**High** (fails to generalize to unseen test data).
Engineering Fixes	• Increase network depth / model capacity. • Add more expressive polynomial features. • Reduce excessive regularization (λ).	• Gather more diverse training data. • Apply L_2 Weight Decay / L_1 Regularization. • Add Dropout layers & Early Stopping. • Prune decision trees / reduce parameters.

🧠 M5 Active Recall & Exam Questions

Q1. Differentiate between Underfitting and Overfitting. What techniques resolve overfitting in neural networks? (8 Marks)

- **Underfitting (High Bias):** Occurs when the model is too simple to capture underlying data trends, yielding high errors on both training and testing datasets.
- **Overfitting (High Variance):** Occurs when the model is overly complex, memorizing random noise and sample idiosyncrasies, leading to near-zero training error but high test error.
- **Techniques to Mitigate Overfitting:**
1. *Regularization (L_1 / L_2):* Penalizes large weight magnitudes in the loss function ($\mathcal{L}_{\text{total}} = \mathcal{L} + \lambda ||\mathbf{w}||^2$).
2. *Dropout:* Randomly deactivates a fraction p of hidden neurons during each training step.
3. *Early Stopping:* Halts training when validation loss stops improving.
4. *Data Augmentation:* Synthetically expands training set size through geometric transformations.